

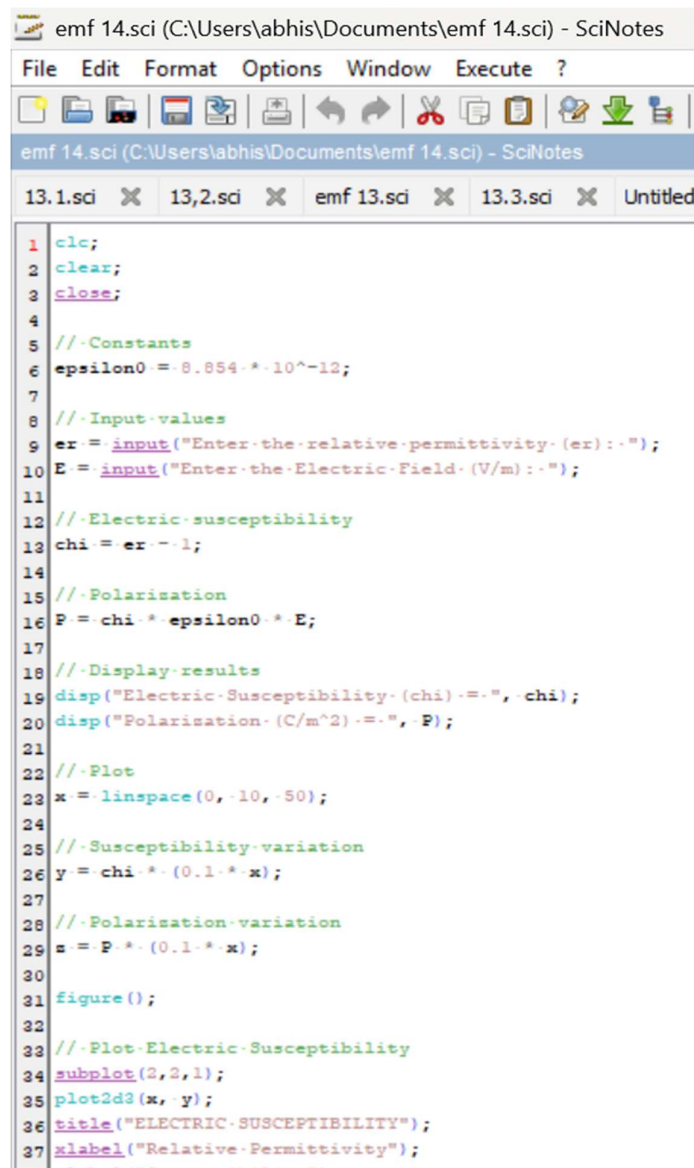
Exp No: 14: Determination and plotting of Polarization in different Dielectric Medium with their relative permittivity and to find electric susceptibility.

Aim: To determine and plot polarization in different dielectric medium with their relative permittivity and to find electric susceptibility.

Software required:

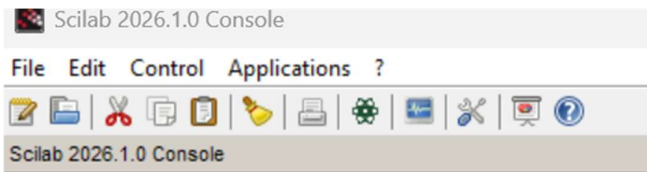
- SCILAB version 6.0.1

Theory: In a dielectric material, when an electric field is applied, the molecules or atoms within the material become polarized. This means that the positive and negative charges within the atoms or molecules shift slightly, creating an internal electric dipole moment. The polarization P is defined as the dipole moment per unit volume of the material and is given by the equation:



```
emf 14.sci (C:\Users\abhis\Documents\emf 14.sci) - SciNotes
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emf 14.sci (C:\Users\abhis\Documents\emf 14.sci) - SciNotes
13.1.sci X 13.2.sci X emf 13.sci X 13.3.sci X Untitled

1 clc;
2 clear;
3 close;
4
5 // - Constants
6 epsilon0 = 8.854 * 10^-12;
7
8 // - Input values
9 er = input("Enter the relative permittivity (er) : ");
10 E = input("Enter the Electric Field (V/m) : ");
11
12 // - Electric susceptibility
13 chi = er - 1;
14
15 // - Polarisation
16 P = chi * epsilon0 * E;
17
18 // - Display results
19 disp("Electric Susceptibility (chi) :- ", -chi);
20 disp("Polarisation (C/m^2) :- ", -P);
21
22 // - Plot
23 x = linspace(0, 10, 50);
24
25 // - Susceptibility variation
26 y = chi * (0.1 * x);
27
28 // - Polarisation variation
29 z = P * (0.1 * x);
30
31 figure();
32
33 // - Plot Electric Susceptibility
34 subplot(2,2,1);
35 plot2d3(x, y);
36 title("ELECTRIC SUSCEPTIBILITY");
37 xlabel("Relative Permittivity");
```



Enter current I (A): 3

Enter radius r (m): 5×10^{-3}

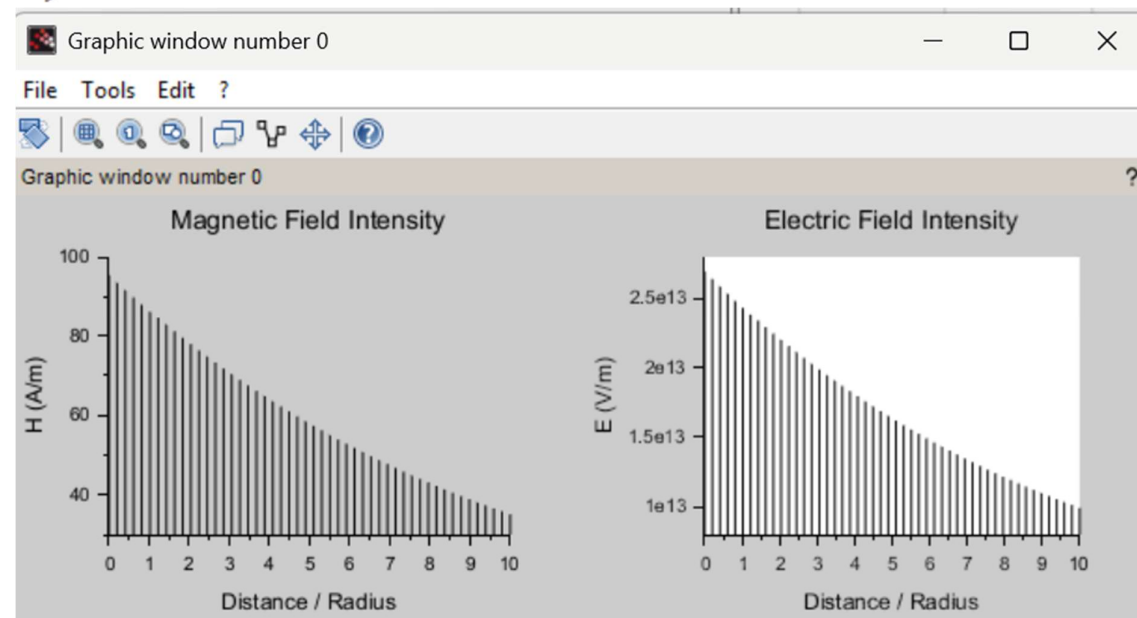
"Magnetic field intensity H (A/m) = "
95.492966

Enter line charge density lambda (C/m): 6

Enter radius r (m): 4×10^{-3}

"Electric field intensity E (V/m) = "
2.696D+13

-->



Finding electric susceptibility polarization

$$\epsilon_r = 6 \times 10^{-6}$$

$$E = 3 \times 10^{-5}$$

Formula:

$$\chi_e = \epsilon_r - 1$$

$$\chi_e = 6 \times 10^{-6} - 1$$

$$\chi_e = -0.999994$$

Result:

$$-0.999994$$

Formula:

$$P = \epsilon_0 \chi_e E$$

$$\epsilon_0 = 8.854 \times 10^{-12}$$

$$P = (8.854 \times 10^{-12}) (-0.999994) (3 \times 10^{-5})$$

$$P = -2.656 \times 10^{-16} \text{ C/m}^2$$

Result:

$$-2.656 \times 10^{-16} \text{ C/m}^2$$

Case 1: To determine and plot the polarization and electric susceptibility of electromagnetic waves in three dielectric media: Air, Water, and Glass, using their relative permittivity.

Analytical: Determine and plot the polarization and electric susceptibility of electromagnetic waves in three dielectric media: Air, Water, and Glass, using their relative permittivity.

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33
clc;
clear;
clf();

//Permittivity-of-free-space
epsilon0 = 8.854e-12; %-----F/m

//Electric-field-strength
E = linspace(0, 1000, 100); %V/m

//Material-properties
materials = ["Air", "Water", "Glass"];
epsilon_r = [1, 80, 6];

//Electric-susceptibility
chi_e = epsilon_r - 1;

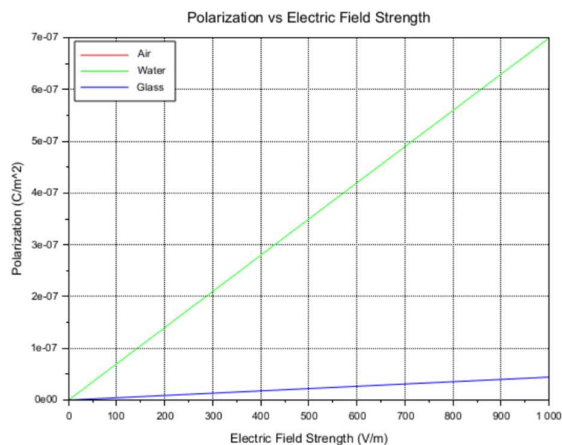
//Calculate-polarisation
P_air = epsilon0 * chi_e(1) * E;
P_water = epsilon0 * chi_e(2) * E;
P_glass = epsilon0 * chi_e(3) * E;

//Plot
plot(E, P_air, 'r', E, P_water, 'g', E, P_glass, 'b');

xlabel("Electric-Field-Strength (V/m)");
ylabel("Polarisation (C/m^2)");
title("Polarisation-vs-Electric-Field-Strength");
legend(["Air", "Water", "Glass"], "in_upper_left");

xgrid();

```



Case 2: To calculate and plot the polarization and electric susceptibility in Silicon, Teflon, and Paper, and understand how their relative permittivity affects the polarization.

Analytical: To calculate and plot the polarization and electric susceptibility in Silicon, Teflon, and Paper

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14.2.sci (C:\Users\abhis\Documents\14.2.sci) - SciNotes
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14.2.sci (C:\Users\abhis\Documents\14.2.sci) - SciNotes
13.1.sci x 13.2.sci x emf 13.sci x 13.3.sci x Untitled 1 x emf 15.sci
14 // Applied electric field
15 E = 1e6; // V/m
16
17 // Calculate polarisation
18 polarisation = epsilon0 * chi_e * E;
19
20 // Display results
21 disp("=====");
22 disp("Material | Relative Permittivity | Susceptibility | Polarisation");
23 disp("=====");
24
25 for i = 1:length(materials)
26     mprintf("%s | %.2f | %.2f | %.6e C/m^2\n", ...
27     materials(i), epsilon_r(i), chi_e(i), polarisation(i));
28 end
29
30 disp("=====");
31
32 // Figure 1: Electric Susceptibility
33 clf(0);
34 clf();
35
36 bar(chi_e);
37 xtitle("Electric Susceptibility for Different Materials", ...
38     "Materials", "Electric Susceptibility (chi_e)");
39
40 xticks([1 2 3], materials);
41
42 // Figure 2: Polarisation
43 clf(1);
44 clf();
45
46 bar(polarisation);
47 xtitle("Polarisation for Different Materials", ...
48     "Materials", "Polarisation (C/m^2)");
49
50 xticks([1 2 3], materials);
51

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Scilab 2026.1.0 Console
File Edit Control Applications ?
Scilab 2026.1.0 Console
=====
"Material | Relative Permittivity | Susceptibility | Polarization"
=====
at line 25 of executed file C:\Users\abhis\Documents\14.2.sci

':': Wrong type for argument 2: Real scalar expected.

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Case 3: To explore how different dielectric materials (Rubber, Oil, and Plastic) affect the polarization and electric susceptibility, and to compare their relative permittivity's.

Analytical: Calculate how different dielectric materials (Rubber, Oil, and Plastic) affect the polarization and electric susceptibility, and to compare their relative permittivity's.

```
14.3.sci (C:\Users\abhis\Documents\14.3.sci) - SciNotes
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13.1.sci X 13.2.sci X emf 13.sci X 13.3.sci X Untitled 1 X emf 15.sci X emf

1 clc;
2 clear;
3
4 //Constants
5 epsilon0 = 8.854e-12; //Permittivity of free space (F/m)
6
7 //Material properties
8 materials = ["Silicon", "Teflon", "Paper"];
9 epsilon_r = [11.7, 2.1, 3.0];
10
11 //Electric susceptibility
12 chi_e = epsilon_r - 1;
13
14 //Applied electric field
15 E = 1e6; //V/m
16
17 //Calculate polarisation
18 polarisation = epsilon0 * chi_e * E;
19
20 //Display results
21 disp("=====");
22 disp("Material | Relative Permittivity | Susceptibility | Polarisation");
23 disp("=====");
24
25 for i = 1:length(materials)
26     mprintf("%s | %.2E | %.2E | %.6E C/m^2\n", ...
27         materials(i), epsilon_r(i), chi_e(i), polarisation(i));
28 end
29
30 disp("=====");
31
32 //Figure 1: Electric Susceptibility
33 clf(0);
34 clf(1);
35
36 bar(chi_e);
37 title("Electric Susceptibility for Different Materials", ...
```

```
Scilab 2026.1.0 Console
File Edit Control Applications ?
Scilab 2026.1.0 Console
=====
"Material | Relative Permittivity | Susceptibility | Polarization"
=====
at line 25 of executed file C:\Users\abhis\Documents\14.3.sci

':': Wrong type for argument 2: Real scalar expected.

-->
```

Case 1: Examine the effect of length of a ferromagnetic material on the magnetic flux density (B) for the following conditions

- ▣ The material is ferromagnetic with relative permeability (μ_r) of 500.
- ▣ The magnetic field intensity (H) is held constant at 100 A/m.
- ▣ The length of the material varies from 0.1 m to 2 m.

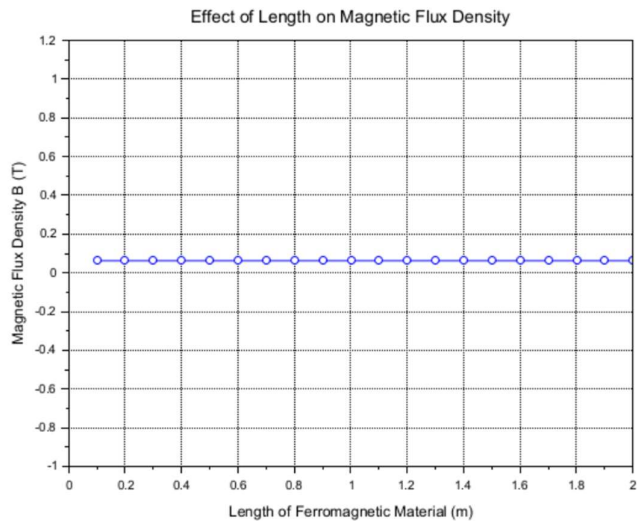
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13.1.sci x 13.2.sci x emf 13.sci x 13.3.sci x Untitled 1 x emf 15.sci

2 clear;
3 clf;
4
5 //Constants
6 mu_r = 500; //Relative permeability
7 H = 100; //Magnetic field intensity (A/m)
8 lengths = 0.1:0.1:2; //Lengths from 0.1 m to 2 m
9 mu0 = 4 * pi * 10^(-7); //Permeability of free space (H/m)
10
11 //Initialize array
12 E_values = zeros(1, length(lengths));
13
14 //Calculate magnetic flux density
15 for i = 1:length(lengths)
16     E_values(i) = mu_r * mu0 * H;
17 end
18
19 //Display results
20 disp("=====");
21 disp("Length (m) ..... Magnetic Flux Density (T)");
22 disp("=====");
23
24 for i = 1:length(lengths)
25     fprintf("%0.1f ..... %0.6e\n", lengths(i), E_values(i));
26 end
27
28 disp("=====");
29
30 //Plot the results
31 plot(lengths, E_values, "-o");
32
33 xlabel("Length of Ferromagnetic Material (m)");
34 ylabel("Magnetic Flux Density B (T)");
35 title("Effect of Length on Magnetic Flux Density");
36
37 //Add grid
38 xgrid();
39
```

Graphic window number 0

File Tools Edit ?

Graphic window number 0



Case 2: To investigate how the magnetic flux density (B) in a ferromagnetic material changes with different widths at a constant length and magnetic field intensity.

Conditions: ☐ Material: Soft iron (same as in Case Study 1)

☐ Length (L): Constant at 10 cm

☐ Magnetic Field Intensity (H): Constant at 1000 A/m

☐ Width (W): Varies from 2 cm to 10 cm (i.e., 2 cm, 4 cm, 6 cm, 8 cm, 10 cm)

```

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13.1.sci X 13.2.sci X emf 13.sci X 13.3.sci X Untitled 1 X emf 1

1 clc;
2 clear;
3
4 // Constants
5 L = 0.1; // Length (m)
6 H = 1000; // Magnetic field intensity (A/m)
7 mu0 = 4 * pi * 10^(-7); // Permeability of free space (H/m)
8
9 // Widths from 2 cm to 10 cm
10 widths = [0.02, 0.04, 0.06, 0.08, 0.10];
11 num_widths = length(widths);
12
13 // Relative permeability of soft iron
14 mu_r = 1000;
15
16 // Initialize array
17 B_values = zeros(1, num_widths);
18
19 // Calculate magnetic flux density
20 for i = 1:num_widths
21     B_values(i) = mu0 * mu_r * H;
22 end
23
24 // Display results
25 disp("=====");
26 disp("Width (m) ---- Magnetic Flux Density (T)");
27 disp("=====");
28
29 for i = 1:num_widths
30     mprintf("%0.2f ----- %0.6f\n", widths(i), B_values(i));
31 end
32
33 disp("=====");
34

```

```

Scilab 2026.1.0 Console
File Edit Control Applications ?
Scilab 2026.1.0 Console

"=====
" Width (m)      Magnetic Flux Density (T) "
"=====
0.02             1.256637
0.04             1.256637
0.06             1.256637
0.08             1.256637
0.10             1.256637
"=====

--> |

```

Case 3: To examine how the magnetic flux density (B) in a ferromagnetic material changes with different values of magnetic field intensity at constant length and width.

Conditions:

Material: Soft iron (same as in the previous studies)

Length (L): Constant at 10 cm

Width (W): Constant at 5 cm

Magnetic Field Intensity (H): Varies from 200 A/m to 2000 A/m (i.e., 200 A/m, 500 A/m, 1000 A/m, 1500 A/m, 2000 A/m)

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14.03.sci (C:\Users\abhis\Documents\14.03.sci) - SciNotes
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13.1.sci X 13.2.sci X emf 13.sci X 13.3.sci X Untitled 1 X emf 15.sci X

4
5 //Define constants
6 L = 0.1; ..... //Length (m)
7 W = 0.05; ..... //Width (m)
8 H_values = [200, 500, 1000, 1500, 2000]; ..... //H (A/m)
9
10 mu0 = 4 * pi * 10^(-7); ..... //Permeability of free space (H/m)
11 mu_r = 1000; ..... //Relative permeability of soft iron
12
13 //Initialise array
14 E_values = zeros(1, length(H_values));
15
16 //Calculate magnetic flux density
17 for i = 1:length(H_values)
18     H = H_values(i);
19     mu = mu0 * mu_r;
20     E_values(i) = mu * H;
21 end
22
23 //Display results
24 disp("=====");
25 disp("H (A/m) ..... Magnetic Flux Density B (T)");
26 disp("=====");
27
28 for i = 1:length(H_values)
29     fprintf("%d ..... %.6f\n", H_values(i), E_values(i));
30 end
31
32 disp("=====");
33
34 //Plot
35 plot(H_values, E_values, '-o');
36
37 xlabel("Magnetic Field Intensity H (A/m)");
38 ylabel("Magnetic Flux Density B (T)");
39 title("Magnetic Flux Density vs Magnetic Field Intensity for Soft Iron");
40
41 xgrid();
```

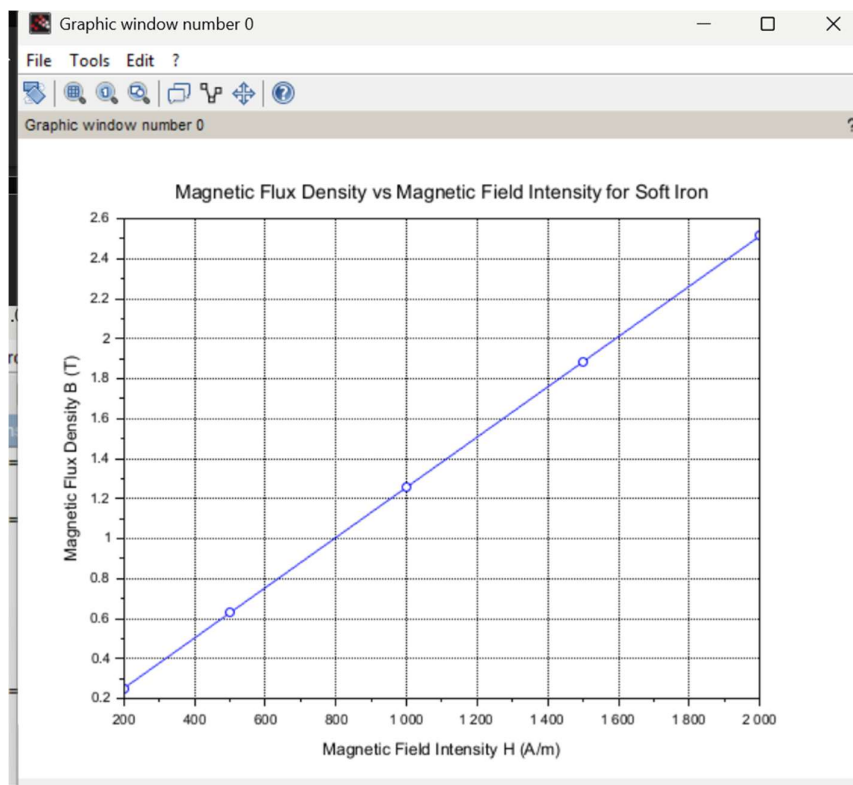
```
Scilab 2026.1.0 Console

File Edit Control Applications ?

Scilab 2026.1.0 Console

=====
" H (A/m)          Magnetic Flux Density B (T) "
=====
200                0.251327
500                0.628319
1000               1.256637
1500               1.884956
2000               2.513274
=====

-->
```



Result : Thus the program was verified for program for Magnetic flux density in ferromagnetic materials using SCILAB was successfully.